

Ruiyan Huang

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EDUCATION

Yale University

Expected May 2026

B.S., Statistics and Data Science; Minor in Climate Sciences and Solutions (GPA: 3.92/4.00)

- **Coursework:** Intermediate Machine Learning, Multivariate Statistics, Causal Inference; Linear Models; Data Structures & Algorithms, Full-Stack Web Development; Intensive Real Analysis

RESEARCH EXPERIENCE

NLP Research Assistant | Yale NLP & Genomics Lab

New Haven, CT · Oct 2025 – present

- Benchmarked methods that translate high-dimensional single-cell RNA-seq data into textual descriptions using large language models; assist in designing model prompts, training/evaluation pipelines, and data preprocessing.

Machine Learning Researcher, Shan Shui Conservation Center

Remote / Tibet (field site) · Apr 2025 – Aug 2025

- Built computer-vision pipeline for individual snow-leopard re-identification from sparse, noisy camera-trap images, combining PIE (hybrid model with splines + XGBoost) with SIFT-based feature matching on top of convolutional neural net embeddings.
- Performed data cleaning, image background subtraction, augmentation, and benchmarked alternative embeddings and distance metrics.

Biostatistics Research Assistant | Yale School of Medicine (Dr. Guan Leying)

New Haven, CT · Mar 2024 – May 2024

- Implemented causal inference and optimization with convex M-estimators on small-sample public health datasets, benchmarked with traditional maximum likelihood estimators.

TEACHING EXPERIENCE

Teaching Assistant, S&DS 312: Linear Models

Yale University · Jan 2025 – present

- Held weekly office hours to support students with problem sets and review sessions reinforcing linear regression, model diagnostics and selection using R and Python.

WORK EXPERIENCE

Data Scientist Intern | Carbon Containment Lab

New Haven, CT · May 2025 – Aug 2025

- Built a bottom-up spreadsheet model for cooling emissions in a U.S. + China commercial real-estate portfolio (2020–2060), including baseline logic and a scenario engine that recomputes emissions and cost curves under alternative grid and adoption pathways.

Climate Data /ESG Analyst Intern | Suzano S.A.

Shanghai, China · Feb 2023 – Aug 2023

- Benchmarked Suzano's ESG performance against major global competitors by analyzing sustainability ratings, climate disclosures, value-chain risks, and governance metrics.

OTHER PROJECTS & PUBLICATIONS

- **Independent Research (Graph Theory):** Simulated triangulations and quivers on non-orientable surfaces with proofs; publication at [Involve Math Journal](#) (2023) and arXiv:2201.05643 (2022).
- **Geospatial Mapping with Supervised Machine Learning:** Processed large-scale geospatial datasets in ArcGIS/R to classify ecological recovery patterns from abandoned oil wells.